



PDAN8411

PROGRAMMING FOR DATA ANALYTICS 1

POE PART 2

Classification and Model Improvement

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Breast Cancer Classification Analysis Report (Part 2)

Introduction

The medical aid scheme asked us to build a model that can quickly identify patients who might need cancer-related benefits, starting with breast cancer. The goal is to tell whether a breast tumour is malignant, which means cancerous, or benign, which is non-cancerous, using measurements from digitised cell images. This report walks through the whole process of choosing a good dataset, exploring the data, preparing it for machine learning, training and improving a classification model, and finally creating a simple function that can give a prediction for a new patient.

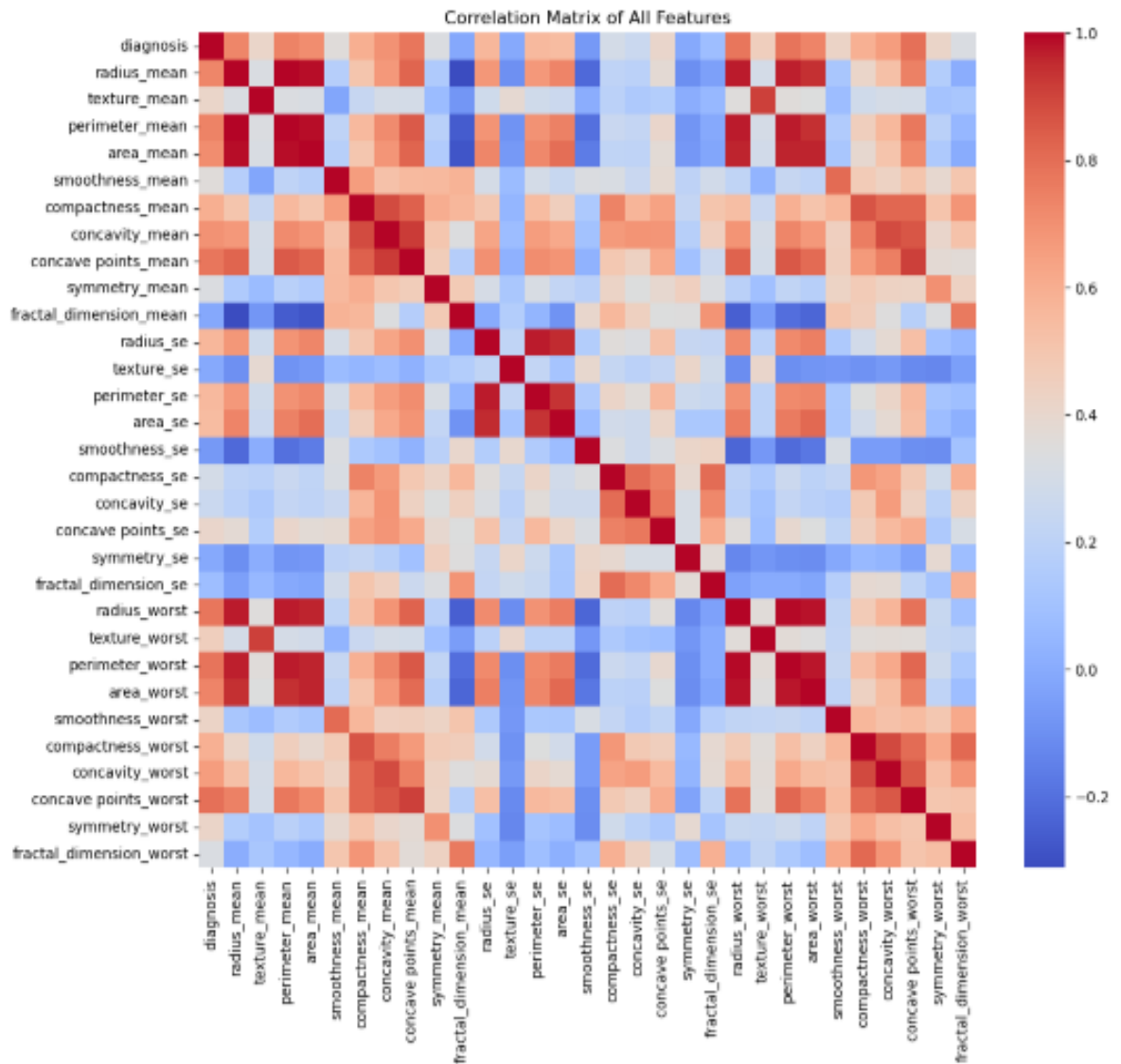
Planning the Analysis

Before writing any code, I made a clear plan to make sure every part of the assignment was covered. The plan includes me justifying why the chosen dataset is suitable for classification and doing an exploratory data analysis to check for missing values and see how the classes are distributed, preparing the data by encoding the target, which would be malignant/benign for breast tumours, and scaling the numeric features, choosing a set of features, which I kept all 30 measurements at first, then later used the model's own feature importance scores to confirm they are useful, comparing three different classifiers for logistic regression, random forest, and SVM using the 10-fold cross-validation, tuning the best model with a grid search, evaluating the final model on a test set that was never used during training, and finally writing a reusable prediction function and this report (Scikit-learn 2026).

Exploratory Data Analysis

The Wisconsin Breast Cancer Diagnostic dataset was gotten from Kaggle and then loaded from a CSV file on my laptop. It contains 569 samples, each with 30 numeric features like mean radius, texture, perimeter, etc. and a column called 'diagnosis' that says whether the tumour is malignant with label 'M' or benign with label 'B'. As these are not useful for the prediction, I took away a column with no data named Unnamed: 32 and an 'id' field. Since there were no missing values in the data, no imputation was required. The data was moderately balanced, the class allocation is 212 malignant and 357 benign. A stratified train/test split helps me to maintain this balance.

I made a few charts to help me see the contrasts in features between benign and malignant tumours. Diagnosis countplots just reveal how many cases fall within each category. Features such as radius_mean coloured by diagnosis' histograms reveal that malignant tumours usually have greater average radii; hence, more erratic cells are more prone to being cancerous. A correlation heatmap reveals how every feature connects to the diagnosis; dark red suggests a significant good association with cancer. For instance, the diagnosis correlates at 0.79 with concave points_worst, indicating that cancer is highly connected to more aberrant cell forms. Characteristics like symmetry_se almost have no relationship at all. As one would expect, the heatmap also reveals that many of the worst_* characteristics are highly correlated with one another, as they all define related physical qualities.



Correlation heatmap: Correlation matrix of the diagnosis (malignant = 1) with all 30 variables. Stronger positive correlations having a darker red mean that in malignant tumours, bigger values of that feature are more common.

Data Processing

Processing of data using the LabelEncoder, the diagnosis column was converted to numbers, whereby the 'malignant' feature became 1 and the 'benign' feature became 0. Target y is the encoded diagnosis; the feature matrix X contains all 30 numerical columns. train_test_split with stratify=y split the data into a training set of 80% of the samples and a test set of 20% of the samples. This guarantees an equal ratio of malignant and nonmalignant cases in both sets. I used a StandardScaler within a pipeline since classifiers such as logistic regression and SVM are sensitive to the range of the variables, which would imply the scaler picks the mean and standard deviation just from the training data and then applies the same transformation to both the training and test data, hence preventing any "data leakage" that would make the model look better than it actually is (Scikit-learn 2026).

Feature Selection

All 30 features were retained instead of manually removing variables because each measurement has medical relevance. After training, the Random Forest model's `feature_importances_` attribute was analysed using a feature importance bar chart to identify the most influential variables. The three most important features were worst concave points, worst perimeter, and worst radius. These features are linked to the size and shape of cell nuclei. This aligns with established medical knowledge, since larger and more irregular cell structures are more commonly associated with malignant tumours.

Model tuning

Based on ROC AUC as the performance indicator, I assessed three classifiers on the training set using 10-fold cross-validation. The average AUCs for logistic regression were 0.9953 (+0.0081), random forest 0.9884 (+0.0185), and SVM 0.9950 (+0.0079), which showed that all three did quite well. I went with Random Forest for more tweaking because it naturally handles non-linear interactions, gives a clear ranking of feature importance, and isn't as likely to overfit as a single decision tree (Scikit-learn 2026).

Tuning Hyperparameter Configurations

I utilised grid search to evaluate these three key parameters in the dataset: the number of trees, the maximum depth for each tree, and the minimum sample size required to divide a node. As this was to make the Random Forest work better. The test used 5 parts to check the training dataset. The best settings found were 50 trees, no limit on how deep each tree can be and 2 samples needed to split a node. The best score achieved was 0.9906. The best model, called `best_rf` was then tested on the test set.

Model Evaluation

On the test set which the model had not seen before the improved Random Forest got an accuracy of 97.4%. It was 100% accurate, in identifying cases and got 92.9% of malignant cases correct. It had an F1 score of 96.3% and a ROC AUC of 99.4%. The results showed that all 72 benign cases were correct which is 100% accuracy. Out of 42 malignant cases 39 were correct and 3 were missed. The report also showed that for cases it got all of them correct and was 96% sure and for malignant cases it got 93% correct and was 100% sure.

Test Set Performance:

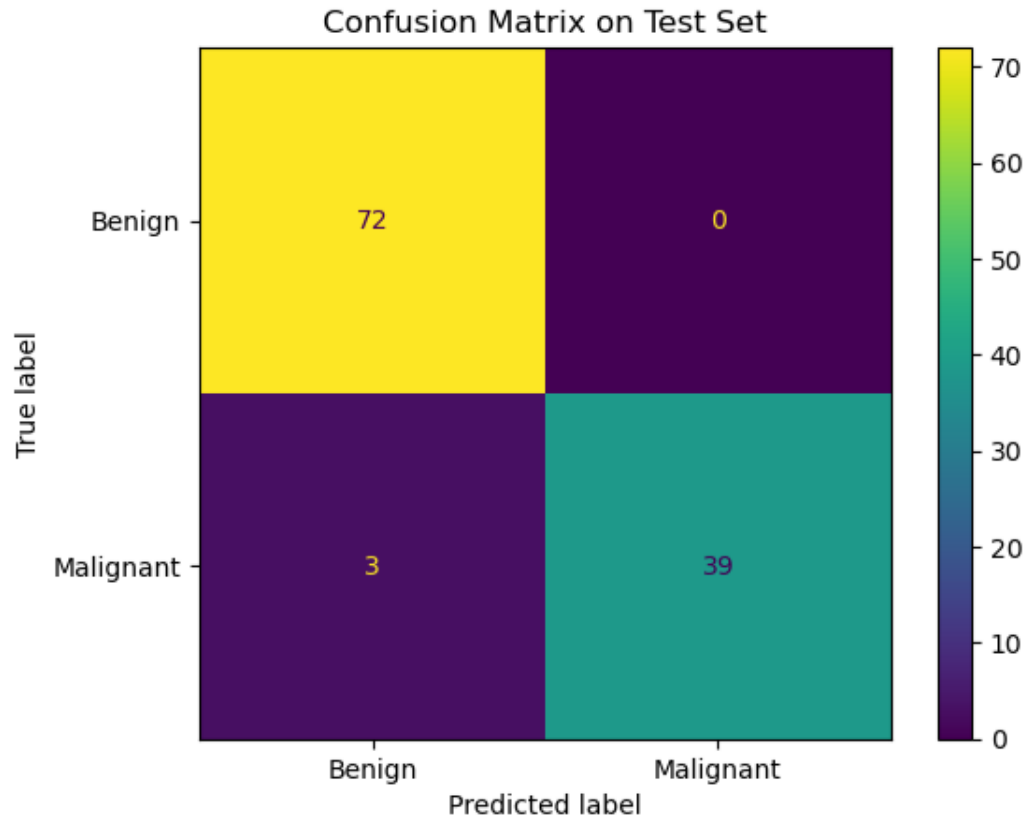
Accuracy: 0.9737

Precision: 1.0000

Recall: 0.9286

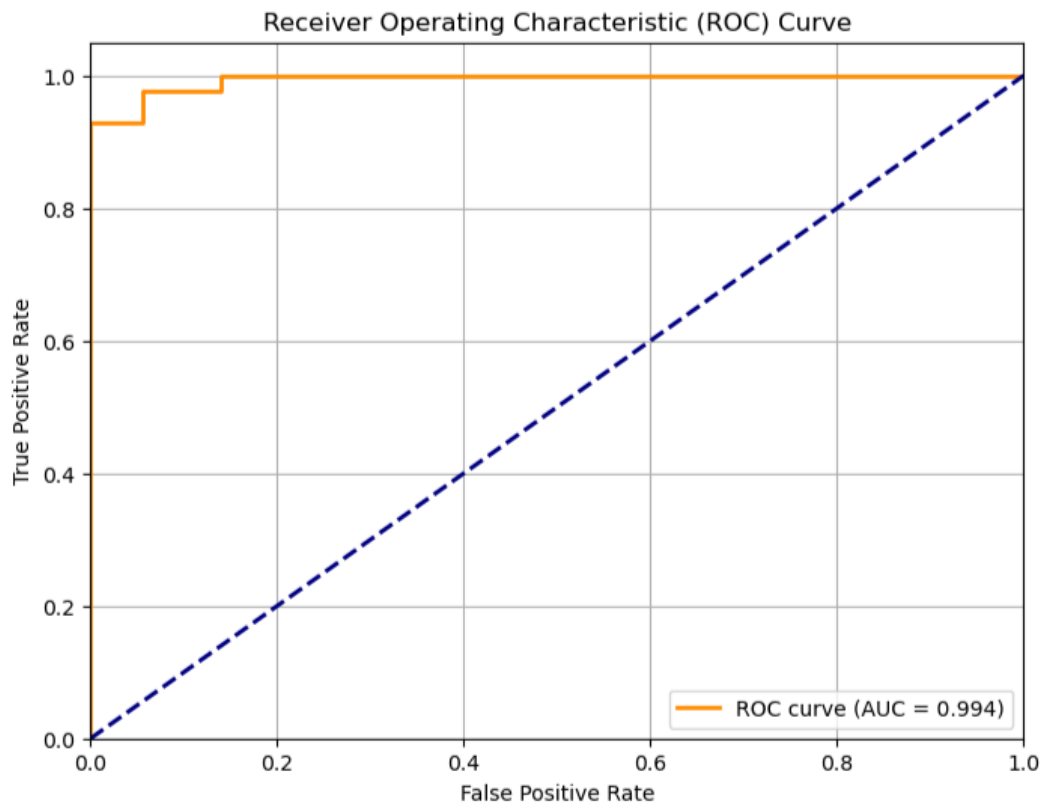
F1 Score: 0.9630

ROC AUC: 0.9940



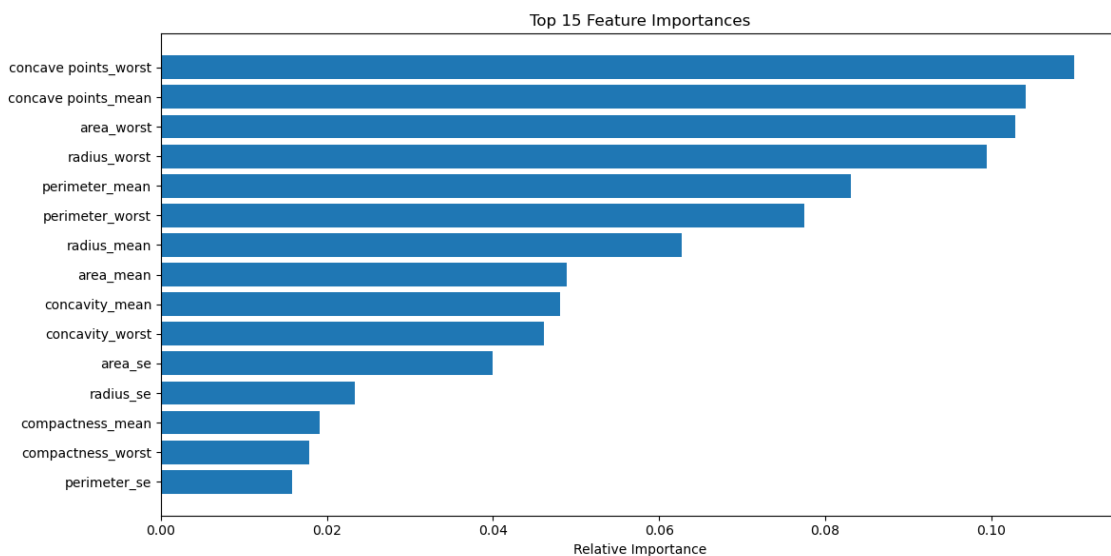
	precision	recall	f1-score	support
Benign	0.96	1.00	0.98	72
Malignant	1.00	0.93	0.96	42
accuracy			0.97	114
macro avg	0.98	0.96	0.97	114
weighted avg	0.97	0.97	0.97	114

The ROC curve plots the true positive rate against the false positive rate as the decision threshold changes. The curve bends sharply to the top-left, and the area under the curve (AUC) is 0.994, this indicates the model is excellent at separating malignant from benign cases.



Findings

The tuned random forest model performs very well, it can correctly identify nearly all benign tumours and about 93% of malignant tumours. The most important features, according to the feature importance chart, are worst concave points, worst perimeter, and worst radius. This makes sense: as more irregular and larger cell nuclei are strong indicators of malignancy, even though Random Forest does not require scaling, I still used a scaler for consistency when comparing it with the other classifiers. The close match between the cross-validation score is 0.9906 and the test AUC is 0.994 which shows that the model generalises well and is not over-fitted.



Model Visualisation

Four visualisations were created to help understand the data and the model. The correlation heatmap gives an overview of how every feature relates to the diagnosis and to each other. The confusion matrix shows exactly where the model made mistakes, three malignant cases were called benign. The ROC curve demonstrates that the model can achieve a very low false-positive rate while keeping a high true-positive rate. The feature importance bar chart tells us which measurements the model relies on most, which can help doctors understand the model's reasoning.

Examining Overfitting

Overfitting is when the model remembers the training data instead of finding general rules. To check this, I compared the validation scores with the test scores. The cross-validation ROC AUC was 0.9906, and the test one for ROC AUC was 0.994 which they are very close. The model is not overfitted. The training accuracy was 100%, but that is normal for a Random Forest model. The small difference is okay.

Execution of a Prediction Model

I made a function called `predict_breast_cancer` to use the model in real life. It takes a list of 30 numbers which is the same measurement as the training set, turns it into a table with the column names, runs the trained model and says either 'Malignant' or 'Benign'. I tried it on the sample from the test set. The model said 'benign', which is correct.

In conclusion

This project successfully built a classification model for breast cancer diagnosis. The Wisconsin dataset is clean and well-suited for the task. The tuned Random Forest model achieves a test ROC AUC of 0.994, an accuracy of 97.4%, and a recall of 92.9%, all strong results for a screening tool. The model is not over-fitted and its decisions can be interpreted through feature importances. A simple prediction function makes it ready for integration into the medical aid scheme's systems.

Recommendations

The scheme can use this model to find patients who're at high risk. This will help doctors because they will not have to spend much time looking at patient information by hand. If the scheme starts collecting its patient information, the model should be updated with this new information, so it stays correct. The scheme should also keep an eye on how the model is working overtime. The scheme needs to pay attention to the times when the model misses patients who have serious problems. The model should be updated whenever the scheme gets information that can be used to make the model better. For better results the scheme could try using a different kind of model that looks at actual pictures. For now the model the scheme is using is good at finding high-risk patients.

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